📘 AI/ML Internship – Day 21 Notes & Tasks

🟡 Module 4: NumPy (Reshaping & Array Manipulation)

📌 Part 1: Why Reshaping is Important?

👉 In AI/ML:

- Data must be in correct shape
- Models expect specific input formats

📌 Part 2: Theory Answers (IMPORTANT)

🔹 1. What is Reshape?

👉 Reshape means changing the structure (shape) of an array without changing its data.

👉 Example:

- 1D → 2D
- 2D → 3D

🔹 2. Rule of Reshape

👉 Total number of elements must remain the same.

✔️ Example:

- 6 elements → reshape into (2 × 3) ✅
- 6 elements → reshape into (4 × 2) ❌

👉 Formula:

Total elements before = Total elements after

🔹 3. Difference Between Flatten and Ravel

Feature	flatten()	ravel()
Copy or View	Copy	View
Memory	Uses more memory	Uses less memory
Speed	Slower	Faster

👉 Simple:

- flatten() → new copy
- ravel() → same data view

🔹 4. What is Transpose?

👉 Transpose means converting rows into columns and columns into rows.

👉 Example:

```
[1 2]      [1 3]
[3 4]  →   [2 4]
```

🔹 5. What is Stacking?

👉 Stacking means combining multiple arrays into one.

👉 Types:

- Vertical stacking (row-wise)
- Horizontal stacking (column-wise)

📌 Part 3: Reshape Array

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 6])

reshaped = arr.reshape(2, 3)
print(reshaped)
```

📌 Part 4: Reshape to 3D

```
reshaped = arr.reshape(1, 2, 3)
print(reshaped)
```

📌 Part 5: Flatten Array

```
arr = np.array([[1, 2], [3, 4]])

flat = arr.flatten()
print(flat)
```

📌 Part 6: Ravel

```
flat = arr.ravel()
print(flat)
```

📌 Part 7: Transpose

```
arr = np.array([[1, 2], [3, 4]])

print(arr.T)
```

📌 Part 8: Stacking

```
a = np.array([1, 2])
b = np.array([3, 4])

print(np.vstack((a, b)))
print(np.hstack((a, b)))
```

📌 Part 9: Splitting

```
arr = np.array([1, 2, 3, 4])

print(np.split(arr, 2))
```

📌 Part 10: Real-World Thinking

👉 In AI:

- Reshape → model input
- Flatten → feature vector
- Stack → combine datasets
- Split → train/test data

🎯 Tasks for Day 21

🧠 Task 1: Theory (Write Answers)

1. What is reshape?
2. Rule of reshape
3. Difference between flatten and ravel
4. What is transpose?
5. What is stacking?

💻 Task 2: Reshape Practice

1. Convert 1D → 2D
2. Convert 1D → 3D
3. Try invalid reshape

💻 Task 3: Flatten

1. Convert 2D → 1D
2. Compare flatten vs ravel

💻 Task 4: Transpose

1. Transpose matrix
2. Verify rows ↔️ columns

💻 Task 5: Stacking

1. Vertical stack
2. Horizontal stack
3. Combine arrays

💻 Task 6: Splitting

1. Split array
2. Split rows